



Roll No:

BTECH
(SEM. IV) THEORY EXAMINATION 2025-26
HYDRAULIC ENGINEERING AND MACHINES

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.**02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define specific energy and critical depth.	CO2	K2
b.	Differentiate between uniform and non-uniform flow.	CO1	K2
c.	Write Manning's equation for open channel flow.	CO1	K1
d.	What is hydraulic jump? State its significance.	CO3	K2
e.	Define specific speed of turbine.	CO5	K1
f.	What are types of surges in open channel?	CO3	K1
g.	What is cavitation in pumps?	CO4	K2

SECTION B

2. Attempt any three of the following:**07 x 3 = 21**

Q no.	Question	CO	Level
a.	Explain Chezy's and Manning's equations for uniform flow.	CO1	K2
b.	Derive condition for most economical trapezoidal channel section.	CO1	K3
c.	A rectangular channel carries discharge of 4.8 m ³ /s with depth 1.6 m. A hump of 0.5 m is introduced. Determine change in water surface level.	CO2	K3
d.	Explain centrifugal pump with neat sketch and working principle.	CO4	K2
e.	Explain cavitation in turbines, its effects and prevention methods.	CO5	K2

SECTION C

3. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain specific energy curve and prove condition for critical flow.	CO2	K4
b.	A rectangular channel (width = 2.5 m) carries 5 m ³ /s at depth 1.5 m. Find specific energy and alternate depth.	CO2	K3

4. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain GVF profiles (M1, M2, S1, S2) with sketches.	CO2	K2
b.	A rectangular channel has slope 0.0009 and depth 1.5 m. Find height of hump to produce critical flow.	CO2	K3



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5. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Derive relation for sequent depth ratio in hydraulic jump.	CO3	K4
b.	A rectangular channel (width = 4 m, discharge = 16 m ³ /s) has depth 0.5 m. Check if jump occurs and find sequent depth & energy loss.	CO3	K3

6. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain reciprocating pump with working and diagram.	CO4	K2
b.	A jet of water strikes a plate moving away with a given velocity. Determine the force exerted, work done, and efficiency.	CO4	K3

7. Attempt any one part of the following:**07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain working of Pelton turbine with neat sketch.	CO5	K2
b.	Design a Pelton wheel turbine for the following data: Power = 11775 kW, Head = 400 m, Speed = 750 rpm, η = 86%. Find wheel diameter and jet diameter.	CO5	K3